

Prashant Singh

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience building scalable backend systems, REST APIs, and AI-driven applications. Hands-on expertise in Python, Django, Angular, and cloud platforms. Proven track record delivering telecom integrations, RAG pipelines, and real-time call infrastructure at production scale.

EXPERIENCE

Software Engineer

Aug 2024 – Present

Python, Django, DRF, PostgreSQL, Twilio, Elastic search, GCP

JAXL

- Integrated **Plivo** for outbound/inbound calling and **Twilio** for voice verification and OTP-based SMS authentication, enabling secure and reliable telecom workflows.
- Built a call tag feature enabling agents to annotate and full-text search calls by custom tags, reducing average call-retrieval time.
- Architected a high-scale **Call Campaign Engine** supporting Manual, IVR, and AI-driven conversational calling modules with decoupled agent execution — enabling independent scaling of each module.
- Investigated and resolved **PostgreSQL index bloat** on the `latest_activity` datetime field — performed RCA, tracing excessive index churn from frequent per-activity updates, and redesigned the update strategy to eliminate dead tuple accumulation and restore query performance.

Software Engineer

Aug 2022 – Aug 2024

Angular, Python, Flask, RAG, Milvus, Faiss, MongoDB, GCP

Infosys

- Built a **Retrieval-Augmented Generation (RAG)** API using Flask and milvus vector storage, enabling semantic search over large enterprise document corpora.
- Contributed to dataset preparation and fine-tuning workflows for LLM projects, spanning data cleaning, prompt engineering, and evaluation.
- Deployed and managed applications on **Google Cloud Platform (GCP)**, improving environment consistency and reducing deployment turnaround.

Software Engineer Intern

Apr 2022 – Jul 2022

Python, Django, Apple Pay

JAXL Innovations

- Enhanced Django Admin with custom filters, search fields, and relational data views to streamline internal data management.
- Defined Django model constraints and wrote unit test cases, increasing test coverage from 60% to 85%.
- Handled Apple In-App subscription lifecycle notifications — `DID.RENEW`, `EXPIRED`, `CANCELLED`, and `SUBSCRIBED` — updating subscription states reliably on each server-side event.

PROJECTS

Vargi-Bot | *ROS, Python, OpenCV, MQTT, JavaScript, Google App Script, HTML/CSS*

- Built a 3D dynamic warehouse simulation on Gazebo for the e-Yantra Robotics Competition (EYRC) as part of a 4-member team.
- Implemented a robotic arm sorting system using ROS Manipulation and OpenCV perception to classify packages by medical priority.
- Integrated MQTT and Google App Script to sync shipping/delivery data to Google Sheets with email push notifications and a live AJAX-updated dashboard.

TECHNICAL SKILLS

Languages: Python, JavaScript, C/C++, HTML/CSS

Frameworks: Django, Django REST Framework, Flask, Angular

Databases: PostgreSQL, MongoDB, Milvus (Vector DB), Redis

Cloud & DevOps: GCP, AWS SageMaker, Jupyter

Telecom: Twilio, Plivo, Exotel

Other: ROS, MQTT, OpenCV, REST APIs, RAG, Machine Learning, LLMs